

Time to complete: 62:29

1. الإسم الرباعي (بالعربي فقط)

0 / 0 pts

Auto-graded

2. رقم الجلوس

0 / 0 pts

Auto-graded

3. المستوي

0 / 0 pts

Auto-graded

☐ الأول☐ الثاني☒ الثالث☐ الرابع 2013☐ الرابع 2014☐ الرابع 2015☐ الرابع 2016☐ الرابع 2017☐ الرابع 2018

0 / 0 pts

Auto-graded

5. الصاله

0 / 0 pts

Auto-graded

☐ أ☐ ب☒ ج☐ د☐ هـ

6. رقم الكمبيوتر

0 / 0 pts

Auto-graded

77

7. الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس).

0 / 0 pts

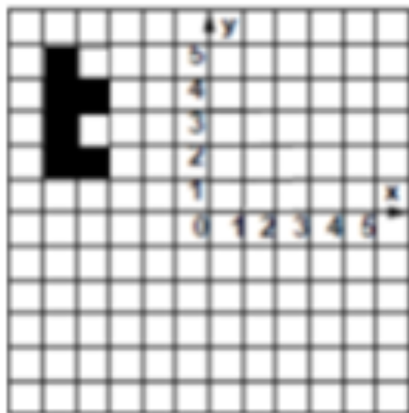
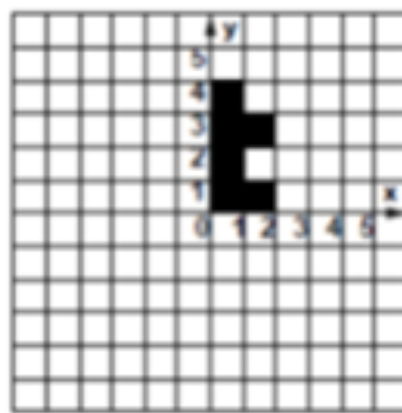
Auto-graded

ZNEY

8. The OpenGL command(s) that makes the transformation from Figure A to Figure B

1 / 1 pt

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**Figure: A****Figure: B**`glTranslatef(5.0, -1.0, 0.0);``glTranslatef(-5.0, 1.0, 0.0);``glRotatef(90.0, 0.0, 0.0, 1.0); glTranslatef(-1.0, 1.0, 0.0);``glTranslatef(-1.0, 1.0, 0.0); glRotatef(90.0, 0.0, 0.0, 1.0);`

9. The RGB of a yellow color is

1 / 1 pt

Auto-graded

(1.0,1.0,0.0)

(1.0,0.0,1.0)

(0.0,1.0,1.0)

(1.0,1.0,1.0)

10. The following two code segments are equivalent:

1 / 1 pt

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Code#1

```
glScalef(-1.0, 1.0, -2.0);
```

```
glScalef(-1.0, 1.0, 4.0);
```

```
dawShape_3();
```

Code#2

```
glScalef(1.0, 1.0, -4.0);
```

```
glScalef(1.0, 1.0, 2.0);
```

```
dawShape_3();
```

True

False

11. Double buffering solves which problem:

1 / 1 pt

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producing stereo images

low memory bandwidth

remote debugging PCs

displaying from a frame buffer while it's being updated

optimizing scheduling two elevators as a group

12. Rotating the 3D point (0,-1,0) by 90 around z-axis gives what:

0 / 1 pt

Auto-graded

☒ (1,0,0)☐ (-1,0,0)☐ (0,1,0)☐ (0,-1,0)☐ (-7,7,0)

13. Find a normal vector to the plane through the points (1,0,0), (1,1,0), (1,0,1)

1 / 1 pt

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☐ (0, -1, 0)☐ (0, 1, 0)☒ (1, 0, 0)☐ (1, 1, 0)☐ (3, 2, 2)

14. This question is about 2D geometry. The matrix for a translation by 2 in x and 3 in y is what?

0 / 1 pt

Auto-graded

A. $\begin{pmatrix} 2 & 0 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ B. $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ C. $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 3 & 1 \end{pmatrix}$ D. $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}$ E. $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}$

☐ A☐ B☐ C☒ D☐ E

15. Normalize this vector: (8,-6,0).

1 / 1 pt

Auto-graded

(3, 4, 5)

(3/5, -4/5, 0)

(4/5, -3/5, 0)

(4/5, 3/5, 0)

(8, -6, 10)

16. The Modelview matrix after executing the following OpenGL code segment is:

1 / 1 pt

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```
glMatrixMode(GL_MODELVIEW);
```

```
glLoadIdentity();
```

```
glRotatef(45.0, 0, 0, 1.0);
```

```
glRotatef(45.0, 0, 0, 1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

A

B

C

D

17. The Modelview matrix after executing the following OpenGL code segment is:

1 / 1 pt

Auto-graded

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glScalef (2.0,-1.5, -3.0);
glRotatef(180.0, 0, 0, 1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} -1 & 0 & 0 & 2 \\ 0 & -1 & 0 & -1.5 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & 1.5 & 0 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1.5 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & -1.5 & 0 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

A

B

C

D

18. Write down the Modelview matrix after executing the following OpenGL code segment.

1 / 1 pt

Auto-graded

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glPushMatrix();
glScalef (-1.0,0.5, -1.0);
glTranslatef(-1.0f, -2.0f, -3.0f);
glRotatef(-90.0, 0, 0, 1.0);
glPopMatrix();
glScalef(-1.0, 2.0,-1.0);
glRotatef(90.0, 0, 0, 1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -2 & 0 & -1 \\ -1 & 0 & 0 & -2 \\ 0 & 0 & -1 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

A

B

C

D

19. If the scaling factor value is less than one, then

1 / 1 pt

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It reduces the size of object

It increases the size of object

It does not change the shape of an object

20. What is the correct order of operations of the OpenGL graphics pipeline?

1 / 1 pt

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projection transformation, modelview transformation, divide by w, viewport transform

modelview transformation, divide by w, projection transformation, viewport transform

modelview transformation, viewport transform, divide by w, projection transformation

modelview transformation, projection transformation, divide by w, viewport transform

21. What is the difference between glColor3d and glColor3f?

0 / 1 pt

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glColor3d only sets RGB, while glColor3f sets R, G, B and A

glColor3d allows 3d color operations, while glColor3f only allows 8-bit

glColor3d takes double arguments, while glColor3f takes float arguments

glColor3d is in real space, glColor3f is in integer space

22. Which function call sets up the size of the output area?

1 / 1 pt

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glViewport()

gluPerspective()

glDisplayfunc()

None of These

23. Does OpenGL use column-major order or row-major order when representing a transformation matrix?

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row-major

column-major

24. The OpenGL function used to set up a parallel view volume is

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glutCreateWindow

glRectf

glOrtho

glViewport

25. What should be done to rotate around a point that is not the origin?

0 / 1 pt

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Translate to origin, rotate about the origin, then translate back to the original position

Rotations can only be performed around the origin

Perform a glRotate and specify the point to rotate around

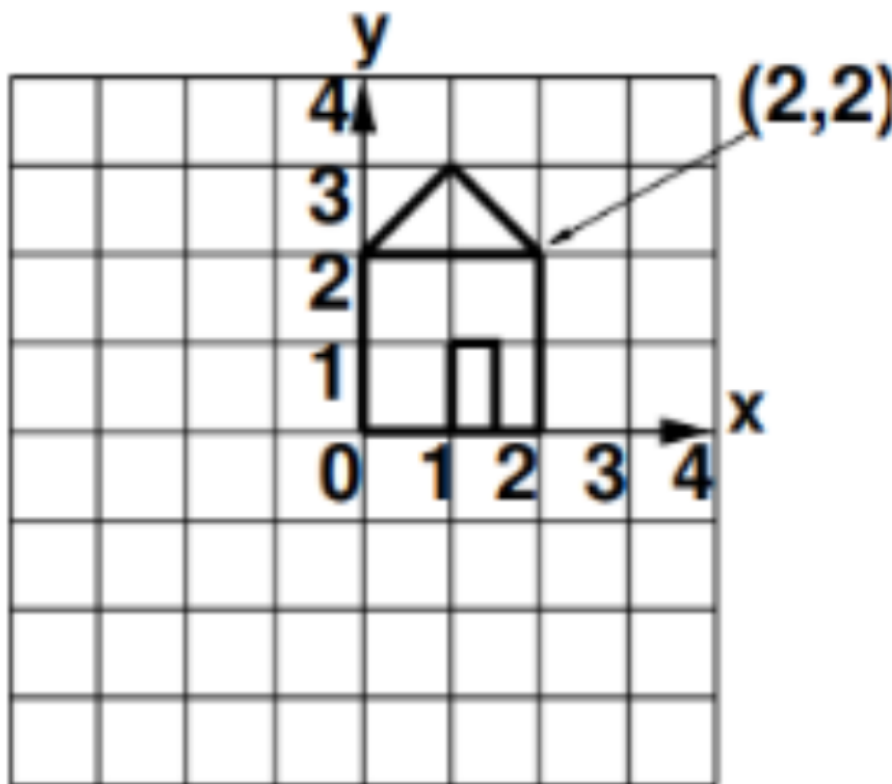
26. Assume that drawHouse function draws the previous house

0 / 1 pt

Auto-graded

```
glMatrixMode (GL_PROJECTION); glLoadIdentity();
glOrtho (-10.0, 10.0, -10.0, 10.0, -2.0, 2.0);
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glTranslate(1, -1, 0);
glRotate(-90, 0, 0, 1);
glRotate(90, 0, 0, 1);
glTranslate(2, 1, 0);
drawHouse();
```

..... is the coordinates of point (2,2) after executing the previous OpenGL code segment



(3,1)

(4,2)

(5,2)

(-2,-2)

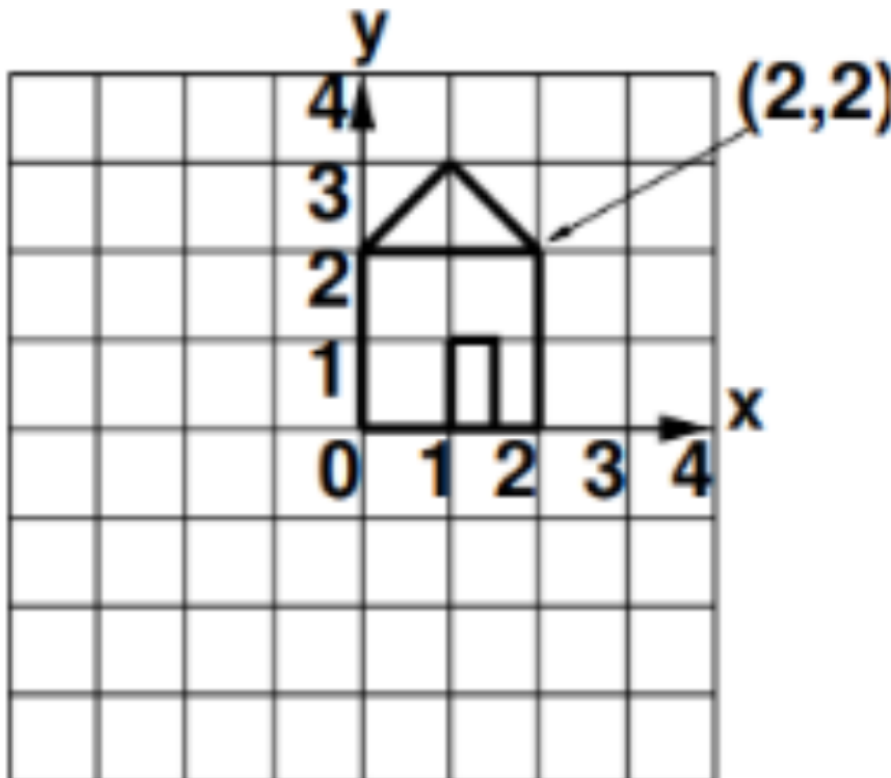
27. Assume that drawHouse function draws the previous house

0 / 1 pt

Auto-graded

```
glMatrixMode (GL_PROJECTION); glLoadIdentity();  
glOrtho (-10.0, 10.0, -10.0, 10.0, -2.0, 2.0);  
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glTranslate(2, 0, 0);  
glScalef(1.0, -1.0,1.0);  
glRotate(180, 0, 0, 1);  
drawHouse();
```

..... is the coordinates of point (2,2) after executing the previous OpenGL code segment:



(0,-2)

(2,2)

(2,0)

(0,2)